Safe Few-Shot World-Model Adaptation to Unseen Joint-Locking Failures

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***Abstract*—Robot world models deployed after hardware damage must adapt to unseen interaction dynamics while exactly respecting the altered mechanism. We study planar pushing with a serial manipulator whose diagnosed joint is locked at a known angle, while friction, delay, payload, and contact effects remain unknown. We introduce a contact-aware physical-context rollout-risk intervention-projected world model (PC-RR-IPWM). It combines analytic lock projection, a frozen contact-aware robot transition, a shared object model, explicit pusher geometry, and a support-aware low-rank intervention residual. An observable physical-context posterior inferred from 25 safe transitions modulates the residual through an exact-zero centered gate and a delayed rollout-risk schedule. Against a compute-, data-, and adapter-matched shared model, object rollout RMSE improves in all 18 combinations of three seeds, two held-out domains, and H10/H25/H50, from 2.04% to 37.84%. Lock violation is zero and controls remain practically equivalent. [REAL-ROBOT TODO: Add real-arm result.]**

I. Introduction

A locked joint changes the robot state manifold discretely, while contact dynamics vary continuously with friction, backlash, delay, and payload. An unconstrained model can violate the known lock; fitting a complete model after failure is data-intensive and unsafe. Fault-tolerant methods adapt behavior [1], [2], and rapid adaptation infers latent properties from recent transitions [3], [4]. Pushing models learn uncertain contact dynamics [5], [6], [7], [8], but typically assume an intact robot or do not enforce the damaged manifold.

We propose PC-RR-IPWM, which assigns known damage to analytic projection and allows unknown physics to modify only an unsupported-intervention object residual. Projected pusher geometry enters the object block through a stop-gradient bridge. A centered context gate gives exact base recovery, and a depth schedule limits accumulated residual error.

Our contributions are: (i) exact projection plus contact-aware block-coordinate dynamics; (ii) a support-aware geometry/latent object intervention; (iii) an observable physical-context gate with reversible fallback; and (iv) a strict three-seed matched evaluation and component ablations.

II. Related Work

Fault adaptation. Damage recovery has been approached by repertoire search [1], adversarial fault training [2], latent extrinsics [3], residual simulator tuning [4], and meta-learned adaptive control [9]. We instead adapt a predictive manipulation model after a diagnosed discrete topology change.

Contact dynamics. Learned forward models support visual and model-based planning [5], [10]. Few-sample GP models [6], hybrid online identification [11], recurrent sim-to-real adaptation [8], and SE(2)-equivariant pushing models [7] address object physics. Our structural prior instead couples exact damaged-robot feasibility to explicit pusher geometry.

AdaptiGraph estimates physical properties online to adapt graph dynamics [12]; PIN-WM identifies a physics-informed rigid-body world model from few visual interactions [13]; and ActivePusher combines residual physics, active data acquisition, and planning [14]. These methods adapt object or environment physics. Our problem also changes the robot's feasible manifold: the lock is diagnosed and exact, while its downstream contact consequences are unknown. Interaction networks and graph simulators encode relational structure [15], [16], and structured world models ground manipulation in human video [17]. Our factorization is intervention-specific rather than a generic graph prior.

Table I is not a global superiority claim. PIN-WM provides a richer physics engine and ActivePusher active planning; our contribution is orthogonal, focusing on the interface between diagnosed robot damage and learned interaction prediction.

III. Problem Formulation

State $s_t = [q_t, \dot{q}_t, o_t]$ contains five joint positions, velocities, and planar object state; $a_t \in \mathbb{R}^5$. Diagnosis gives lock $d = (m, \bar{q})$. Unknown physical context $z \in \mathbb{R}^8$ is inferred from K safe transitions. D3 is held out from D1/D2/D4/D5 training support and is combined with held-out physics. We minimize autonomous H10/H25/H50 error while requiring $q_t^j = \bar{q}^j, \dot{q}_t^j = 0$ for locked coordinates.

We distinguish topology, physical, and compositional generalization. Topology holds out the locked-joint identity; physical generalization changes continuous residual factors; composition combines both. Calibration and evaluation trajectories are disjoint.

IV. Method

A. Analytic projection

$$\Pi_d(q)_j = (1 - m_j)q_j + m_j\bar{q}_j, \quad (1)$$

$$\Pi_d(\dot{q})_j = (1 - m_j)\dot{q}_j, \quad \Pi_d(a)_j = (1 - m_j)a_j. \quad (2)$$

Projection is applied before and after prediction, giving exact feasibility.

TABLE I
SCOPE OF REPRESENTATIVE ADAPTIVE DYNAMICS AND DAMAGE-RECOVERY WORK. HARD CONSTRAINT MEANS FEASIBILITY IS GUARANTEED BY CONSTRUCTION.

Method	Robot damage	Few-shot	Contact model	Hard constraint	Held-out lock
Cully et al. [1]	yes	search	no	no	repertoire
Fault-aware RL [2]	yes	no	no	no	randomized
RMA [3]	wear	yes	no	no	no
TuneNet [4]	no	yes	yes	no	no
AdaptiGraph [12]	no	yes	yes	no	no
PIN-WM [13]	no	yes	yes	physics	no
ActivePusher [14]	no	active	yes	no	no
PC-RR-IPWM	joint lock	yes	yes	yes	yes

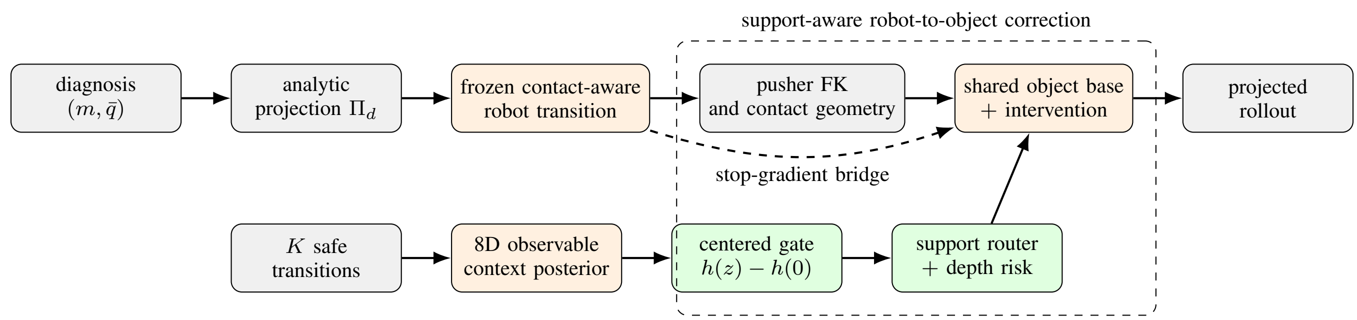

Fig. 1. PC-RR-IPWM deployment graph. Known damage is enforced analytically. The context-gated geometry and latent residual is enabled only outside the training intervention support; K0 recovers the base exactly.

B. *Contact-aware block-coordinate model*

A recurrent robot block predicts free-joint transitions from projected state, action, and contact/object context. Contact input is retained because deleting it destabilizes free-joint pushing rollouts. Forward kinematics produces pusher geometry. A shared object base receives detached robot features, preventing object loss from rewriting the frozen robot block.

The robot block is initialized from the same-seed shared checkpoint. Lock mask and angle do not enter shared input columns that were never trained to consume them; they act through projection. The shared object base is retained and only compact bridge/intervention modules are trained. The implementation is block-coordinate in optimization but contact-aware in its forward pass; we do not claim strict forward block-triangular causality.

C. *Support-aware intervention*

For training lock support $\mathcal{S}$,

$$\nu(d) = \nVdash[m \notin \mathcal{S}]\nVdash[\|m\|_1 > 0].$$

The intact term forces bypass. Object prediction is

$$\hat{o}_{t+1} = f_o(o_t, \mathrm{sg}(h^r_{t+1})) + \nu(d)\alpha_t(z)(r_g + r_\ell),$$

where r_g uses explicit pusher displacement/contact geometry and r_ℓ is a low-rank latent residual.

The geometry branch receives pusher position, predicted displacement, relative pusher–object displacement, and a smooth contact gate. The latent branch captures effects absent from this representation. A bridge aligner maps frozen robot features to coordinates expected by the object head. Leave-one-intervention-out meta-training treats each seen lock as pseudo-unseen; D3 is never observed.

D. *Context and rollout risk*

An amortized posterior estimates 8D context from state/action transitions without simulator labels. The frozen rule uses K25 and scale 1.38. Centering $g(z) = h(z) - h(0)$ guarantees exact K0 recovery. Residual modulation has an eight-step grace period and $\rho(t) = 0.06 \max(0, t-8)$. Support validation and hysteresis reject harmful updates. Conformal intervals quantify context error, not rollout-failure probability.

Posterior output is normalized by fixed per-dimension simulator scales. Dimension- and budget-specific conformal radii are fit on development probes. This separates uncertainty about estimated physical context from risk that an update harms a rollout. The former has calibrated coverage; the latter is controlled by paired support validation and fallback. Posterior variance alone never accepts an update. The depth coefficient is domain-label-free: the shared base dominates early local predictions, then bounded context correction grows only after accumulated shift becomes relevant.

E. *Training and deployment*

Base checkpoints are frozen before intervention training. We optimize

$$\mathcal{L} = \sum_{H \in \mathcal{H}} w_H(\lambda_q \mathcal{L}^H_q + \lambda_o \mathcal{L}^H_o + \lambda_p \mathcal{L}^H_p) + \lambda_T \mathcal{L}^{H_{\max}}_o. \quad (3)$$

The terminal term exposes H50 divergence hidden by mean-over-depth loss. Feasibility is not learned from a penalty. At

TABLE II
FROZEN COMPONENTS AND DEPLOYMENT PERMISSIONS.

Component	Source	Deployment
Robot block	shared/Z69	frozen
Projection	diagnosis	fixed
Object base	shared	frozen
Bridge aligner	meta-train	frozen
Geometry residual	meta-train	context gated
Latent residual	meta-train	context gated
Context posterior	Z65	inference only
Rank-8 adapter	matched support	optimized

deployment, diagnosis supplies $m, \bar{q}$ and one nested safe sequence supplies context. Both models receive identical adapter budgets; support validation accepts the update or returns exact K0. No simulator parameter, domain label, or test trajectory is used.

F. Design rationale and frozen inference

Four negative results constrain the architecture. Topology-conditioned monolithic models did not reliably beat a capacity-matched shared model. Latent optimization and history/FiLM residuals did not produce stable few-shot gains. Independent robot/object factorization failed parameter- and compute-matched tests. Finally, deleting object/contact input from the robot transition degraded long-horizon free-joint prediction. The final design therefore uses a narrow intervention residual around the strongest contact-aware shared base.

The support router prevents the residual from relearning generic object dynamics while degrading intact or seen locks. It reads only membership of the diagnostic mask in the frozen meta-training support, never a test-domain name. Deployment then: (1) fixes diagnosis and projection; (2) collects one nested 25-transition safe sequence; (3) infers context; (4) fits matched rank-8 adapters with identical budgets; (5) accepts or rejects through support validation; and (6) applies projection and the frozen depth ramp at every rollout step. No method-specific extra interaction is permitted.

G. Implementation details

Robot and baseline hidden size is 136. The geometry residual has rank 16, the latent intervention rank 32, and the context modulation rank 16. The context encoder hidden size is 96 and outputs an 8D mean and uncertainty. Residual training freezes the robot block, uses 60 object epochs, learning rate 3×10^{-3} for the final context gate, rollout horizon 50, validation every two epochs, group-robust weight 0.5, and terminal H50 weight 5.0. The matched online adapter performs 12 optimization steps with initial step size 0.2, backtracking factor 0.5 for at most eight steps, trust radius 0.5, and ℓ_2 coefficient 10^{-3}. These values are shared across seeds and are fixed before confirmation.

V. EXPERIMENTS

Both methods use identical training data, K25 support, rank-8 adapters, optimization, and rollouts. The baseline is shared dynamics plus analytic projection and its matched adapter. Seed27 was untouched until the rule froze.

TABLE III
EVALUATION AXES AND EVIDENTIAL ROLES.

Axis	Setting	Question
Topology	D3 held out	Unseen lock?
Physics	held-out mix	Unknown residual?
Budget	K0/K25	Adaptation effect?
Depth	H10/25/50	Stability?
Controls	intact/D2/D4	Negative transfer?
Seed	7/17/27	Rule consistency?

TABLE IV
OBJECT RMSE IMPROVEMENT (%) OVER THE MATCHED BASELINE.

Seed	Domain	H10	H25	H50
7	composition	12.27	5.33	37.84
7	mixed-unseen	15.89	32.50	36.60
17	composition	6.53	13.54	12.52
17	mixed-unseen	9.10	15.50	2.95
27	composition	7.62	2.04	18.98
27	mixed-unseen	9.19	2.85	2.58

A. Protocol, metrics, and fairness

Training includes D1/D2/D4/D5 and variations in friction, backlash, motor strength, delay, and payload. Evaluation combines D3 with held-out residual composition and mixed-unseen interaction distributions. Comparisons pair identical initializations and actions.

Free-joint RMSE excludes the fixed coordinate; object RMSE is primary; pusher RMSE measures task-space propagation. Improvement is

$$100(\mathrm{RMSE}_{\mathrm{shared}} - \mathrm{RMSE}_{\mathrm{ours}})/\mathrm{RMSE}_{\mathrm{shared}}.$$

We freeze a 2% per-cell object gate, 5% free non-regression, zero lock violation, and practical control tolerances. The baseline uses the same projection, K25 support, rank-8 adapter, optimizer, and rollout budget.

The baseline is stronger than an unconstrained predictor because it receives the same lock projection. It asks whether intervention modeling adds prediction value beyond feasibility. K0 isolates context; intact/seen controls test collateral damage; geometry/latent/depth ablations test whether one extra branch alone explains the gain.

All 18 cells exceed 2%; the minimum confirmation is 2.0430%. Locked violation is zero. One seed7 mixed-unseen H50 cell has 5.97% free-joint regression, narrowly missing the frozen 5% gate; all other cells satisfy it. Intact/D2/D4 object differences are at most 1.72%; maximum IID pusher difference is 1.081 mm. K curves are not monotonic, so we claim a frozen K25 rule only.

B. Confirmation and interpretation

Seed27 was opened after scale, grace, and depth ramp froze on seeds7/17. All six confirmation cells exceed 2%, with a

TABLE V
MATCHED SAFETY AND CONTROL CHECKS.

Check	Worst difference
Intact/D2/D4 object	1.72% absolute
Free-joint regression	5.97% worst (one gate miss)
IID pusher position	1.081 mm
Locked violation	0
Context conformal MACE	0.0289

TABLE VI
SEED17 H50 IMPROVEMENT (%) OVER MATCHED SHARED.

Variant	Composition	Mixed-unseen
K0 matched adapter	-122.55	-96.62
K25, no depth risk	27.57	-3.86
Full K25	12.52	2.95

2.0430% minimum. With three seeds we do not claim population significance; we report every cell and claim directional consistency under the frozen protocol.

An earlier projected-only comparison had one 1.9579% cell. It remains a development near-miss but is superseded by the strict matched-adapter decision. The result was not rounded or retuned; the final protocol removed an adapter mismatch.

C. Why K25 closes the long horizon

Context reverses the composition failure but still overshoots one mixed-unseen rollout. Depth scheduling sacrifices some composition gain while making the remaining negative cell positive. The frozen gate optimizes all-cell reliability, not maximum mean improvement. K5/K10/K50 curves are supportive but non-monotonic.

Removing intervention reduces mixed-unseen gain to at most 0.41%. In composition, removing geometry reduces H10 from 6.53% to 0.08%; removing intervention yields 0.01/0.15/1.35% at H10/H25/H50. Removing final projection creates nonzero locked error, whereas retaining it guarantees feasibility. Geometry supplies a direct path, latent correction models remaining physics, context selects correction, and depth risk limits accumulation.

D. Reproducibility and compute

Frozen YAML files define seeds, splits, adapters, and gates. Tracked aggregate JSON backs the reported values; per-window rows and paired uncertainty intervals remain to be exported in the final audit package. The current source of truth is matched_gate_summary_v2.json. The evidence-freeze repository has 244 passing tests. GPU PyTorch handles training, while MuJoCo collection and Python rollout dominate serial time. Identical caches prevent a compute/data fairness confound.

Failed runs and protocol changes remain in a failure ledger. Seed IDs determine data, initialization, and evaluation queries. Confirmation artifacts are immutable after opening. Tables are generated from the same JSON consumed by tests, reducing transcription risk; parameter counts, checkpoint identities, configuration hashes, and timing metadata accompany runs.

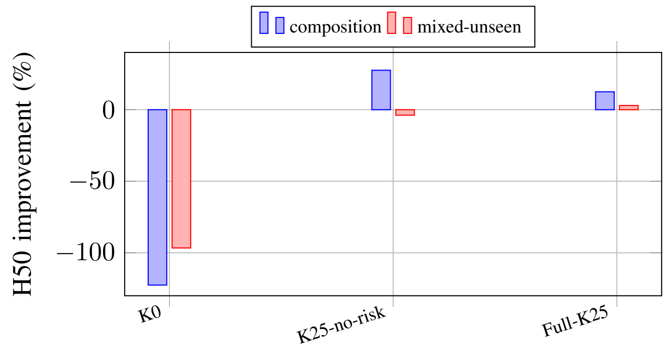

Fig. 2. Seed17 long-horizon mechanism. Context reverses catastrophic K0 drift; depth risk trades peak gain for an all-positive frozen gate.

TABLE VII
SEED17 D3 MIXED-UNSEEN OBJECT IMPROVEMENT (%).

Variant	H10	H25	H50
Full	9.10	15.50	2.95
No geometry	4.57	12.36	1.17
No latent	3.54	2.50	3.32
No intervention	0.01	0.09	0.41
No depth risk	3.14	13.98	-3.86

E. Claim matrix

Evidence supports: C1, projection guarantees zero lock violation; C2, complete K25 exceeds the matched baseline in every primary held-out object cell; C3, support routing keeps controls within margins; and C4, ablations demonstrate complementary roles. It does not establish population-level significance, monotonicity in K, or calibrated probability of rollout failure.

F. Threats to validity

Baseline leakage. Both systems share initialization and data but train separate matched adapters; small control differences are reported rather than called exact equality. *Selection.* Seeds7/17 are development and seed27 is confirmation only. *Metric mismatch.* Prediction RMSE does not prove task success, so the real-arm table also reports pushing success and aborts. *Simulator fidelity.* G0 calibrates kinematics and safety, while contact physics remains randomized. *Small denominators.* Absolute pusher millimeters accompany relative differences.

G. Real-arm protocol

[REAL-ROBOT TODO: Insert setup photo, calibration diagram, and results.] The frozen protocol uses a five-DoF ST3215 arm, eye-in-hand tracking, 10 Hz, intact/D2/D3, K=0/5/10/25, and three repetitions. Stops trigger on stale vision, feedback timeout, 50°C, raw current 400, or 3.5° lock drift.

REAL-ARM FIGURE PLACEHOLDER
ST3215 arm + eye-in-hand camera + tagged block
D2/D3 lock and table/world frames

Fig. 3. Real setup. [REAL-ROBOT TODO: Replace with one labeled photograph with calibration insets.]

TABLE VIII
REAL-ARM PUSHING. [REAL-ROBOT TODO: POPULATE AFTER COLLECTION.]

Lock	Method	Obj.H10	Obj.H25	Success
Intact	Shared/ours	–	–	–
D2	Shared/ours	–	–	–
D3	Shared/ours	–	–	–

G0 measurements provide servo zero positions, directions, 5°/s limits, link lengths, and emergency torque-disable. D2/D3/D4 holds accumulated 78 cycles and 687 s; the clean final run had at most 0.79° drift and the largest reversal backlash was 3.08°. These are safety margins, not task evidence. The remaining study uses a rigid tagged block and calibrated eye-in-hand camera. Intact checks coordinates, D2 tests a seen intervention, and D3 tests the held-out intervention, each with three repetitions and disjoint calibration/test pushes.

H. Pre-registered real-arm analysis

The primary comparison is D3 object RMSE at H10/H25 under K25. Secondary outcomes are pusher/free-joint RMSE, success, adaptation time, peak current, temperature, lock drift, and aborts. Intact and D2 are non-regression controls. Every trial resets the object and uses disjoint test actions. Per-repetition values and bootstrap intervals are retained, including failed/aborted runs. Scale 1.38, eight-step grace, and ramp 0.06 are not tuned on real outcomes.

The camera supplies object pose in a fixed table frame. Servo current is a safety signal, not calibrated force. If H50 exits the visible or safe workspace, it is marked infeasible rather than shortened post hoc. Thus the real experiment tests a frozen simulation claim instead of becoming another development set.

I. Expected real-arm failure modes

Eye-in-hand occlusion can remove the object during contact; stale poses abort rather than being interpolated. Backlash can cause task-space pusher error even with small encoder drift, so both errors are retained. A software-held lock may draw current without moving, hence temperature/current stops are independent of prediction error. Table friction may lie outside simulation randomization; an adapter must still pass support validation. Communication loss invokes sequential torque disable and remains an abort in the ledger.

VI. LIMITATIONS AND CONCLUSION

Current evidence is predictive, not closed-loop control. Diagnosis must provide lock identity and angle. Contact awareness precludes strict triangular causality. Tests cover single locks and planar pushing. Three seeds support consistency but not a broad population claim. Context coverage does not certify rollout risk, and deployment depends on diagnosis and camera state estimation. Nevertheless, PC-RR-IPWM improves every held-out matched cell while preserving exact feasibility and controls. The defensible result is selective improvement on an unseen intervention under a fair protocol, with analytic safety and reversible fallback. [REAL-ROBOT TODO: Add measured D3 effect, failure modes, and final sim-to-real conclusion.]

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